Vantage Robotics Product Line — Engineering Knowledge Base

Cross-Team Reference | Facts are deliberately scattered across sections | Rev. 3

This knowledge base captures facts about the Vantage Robotics product line as they were contributed by different engineering teams over time. Facts relevant to any single question are frequently spread across multiple, non-adjacent sections — this mirrors how real documentation accretes and is the specific property this document is designed to test: correctly answering many of the questions below requires retrieving and combining facts from two or more separate sections.

Section 1: Product Overview

Vantage Robotics currently ships three product lines: the Sentry line (fixed security robots), the Courier line (indoor delivery robots), and the Atlas line (outdoor mapping robots). Product A in this knowledge base refers to the Sentry-S2 model specifically.

Product A (Sentry-S2) uses Engine X for its primary drive system.

Section 2: Warehouse Logistics Notes

This section is unrelated to product engineering and covers warehouse shipping logistics for finished units. Vantage ships finished units from its Reno, Nevada distribution center, with a standard 5-7 business day ground shipping window for domestic orders.

Section 3: Battery Systems

All Vantage products use one of two battery chemistries: Battery Type L (lithium iron phosphate, used in Sentry and Courier lines) or Battery Type N (nickel-metal hydride, used only in older Atlas-1 units, now discontinued). Battery Type L has a rated cycle life of 2,000 charge cycles before capacity drops below 80% of original.

Section 4: Drive System Engineering

Engine X is a brushless DC drive motor rated at 350W continuous output, originally developed for the Courier line and later adopted by Sentry. Engine X requires a dedicated GPU co-processor for real-time obstacle avoidance path planning — without it, the motor controller falls back to a slower, CPU-only planning loop that increases minimum obstacle clearance distance from 15cm to 45cm.

Section 5: Firmware Release Notes

Firmware version 4.2.1 (current stable) added support for over-the-air updates to the GPU co-processor firmware referenced in Section 4, decoupling GPU firmware updates from full system firmware updates for the first time.

Section 6: Customer Support FAQ Excerpt

"My robot beeps three times on startup and won't move." This symptom indicates the GPU co-processor failed its self-test. Since Engine X (Section 4) requires this co-processor for full-speed operation, affected units should be power-cycled once; if the beep pattern persists, the unit should be returned under warranty rather than operated in CPU-only fallback mode long-term, since sustained CPU-only operation was found to increase drive motor thermal wear in field testing.

Section 7: Warranty Terms

Standard warranty covers manufacturing defects for 24 months from date of purchase. Battery packs are covered separately for 12 months or 500 charge cycles, whichever comes first — notably shorter than the 2,000-cycle rated life mentioned in Section 3, since battery degradation within rated life is considered normal wear, not a defect.

Section 8: Regulatory Certification

All products using Engine X (Section 4) are certified under IEC 61508 SIL 2 for functional safety, contingent on the GPU co-processor obstacle avoidance path being active. Units operating in the CPU-only fallback mode described in Section 4 are explicitly excluded from this certification and should not be deployed in safety-critical zones while in that mode.

Section 9: Facility and Sourcing Notes

The GPU co-processor referenced throughout this document (Sections 4, 5, 6, and 8) is sourced from a single third-party supplier under an exclusive multi-year contract. A supply disruption from this supplier in Q2 caused a documented six-week delay in Sentry-S2 production, unrelated to the shipping logistics window described in Section 2.

Section 10: Courier Line Specifics

The Courier-C1 model also uses Engine X (see Section 4 for full specification) but pairs it with Battery Type L (Section 3) in a larger dual-pack configuration for extended indoor delivery routes, roughly doubling usable runtime relative to the Sentry-S2's single-pack configuration.

Section 11: Atlas Line Specifics

The Atlas-2 outdoor mapping robot does NOT use Engine X — it uses a separate, higher-torque Engine Y drive system rated for rough terrain, and therefore is not subject to the GPU co-processor requirement, the associated IEC 61508 SIL 2 certification described in Section 8, or the beep-pattern startup diagnostic in Section 6, all of which are specific to Engine X.

Section 12: Maintenance Schedule

Recommended maintenance interval for Engine X-equipped units (Sentry-S2 per Section 1, and Courier-C1 per Section 10) is every 1,000 operating hours, including a GPU co-processor connector inspection given the startup-beep failure mode described in Section 6. Atlas-2 units, using Engine Y (Section 11), follow a separate 1,500-hour interval with no co-processor inspection step, since Engine Y has no co-processor dependency.